# Interstellar flight profiles and observational signatures of positive-energy warp shells

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**Abstract**

Accelerating warp shells built entirely from positive-energy matter were recently shown to exist: the radiative momentum warpshell of Le [1] steers by photon-rocket recoil under a closed-form cost law and an acceleration–compactness frontier. What that construction does not contain is a flight plan. A systematic search (a citation scan, a full-text survey of the paper, and an inspection of the author's public reproduction code) found no published computation of interstellar routes under the shell's admissibility constraints. We compute that catalog. Under the full constraint stack, a rigorous ceiling, a numerically mapped frontier band, and a provisional rigorous floor, we obtain fuel–time tables for Proxima Centauri, TRAPPIST-1, Sgr A*, and M31; the crew-lifetime contour ($\tau = 50$ yr) requires cruise rapidity $\eta = 11.53$ for an M31 flyby at mass ratio $m_f/m_0 = 9.5 \times 10^{-16}$. Time-optimal profiles ride the moving frontier, and the operative constraint switches from the thin-shell frontier to thick-wall realizability at compactness $x^* \approx 0.54$. The same integrals yield observational signatures: a saturated accelerating shell is exactly dark toward its destination, and announces arrival with a deceleration flash amplified by $\delta^4 = e^{4\eta}$ and compressed in arrival time, following the general decay law $F \propto e^{7\eta - 3\Delta\eta_{\mathrm{tot}}}$ (for the M31 arrival, $e^{7\eta-72}$, with picosecond-scale structure), while remaining gravitationally silent on the Damour dipole. All numbers are machine-injected from certified catalogs (306 catalog rows and 567 signature rows, dual-path verified with zero omissions); the verification record includes six cross-validations against the author's code and one reproducibility gap, which we state as found.

## 1 Introduction

The warp-drive literature has split into two lines that have not met. One line proves what a shell can be. Alcubierre's metric [4] required negative energy, and the no-go results of Santiago, Schuster, and Visser [5], together with the quantum-inequality bounds of Pfenning and Ford [6] and the eigenvalue analysis of Lobo and Visser [7], established that the defect is generic for metric-first designs. The positive-energy soliton of Lentz [8] did not survive scrutiny: the Eulerian energy density of the zero-vorticity soliton integrates to zero over each slice [5], and a direct Eulerian-frame computation exhibits the negative regions [9]. Source-first constructions then produced genuinely positive-energy shells, static or at constant velocity: the taxonomy of Bobrick and Martire [10] and the numerically verified constant-velocity shells of Fuchs *et al.* [11], built with the Warp Factory toolkit [12]. Le closed the acceleration gap: the radiative momentum warpshell [1] wraps a flat passenger cavity in a timelike shell matched to the exact Kinnersley photon rocket [15], satisfies the dominant energy condition in bulk and on the shell, and steers under the closed-form control law $-\dot{m} \geq 3m|a|$ with a Tsiolkovsky budget $m_f/m_0 = e^{-3\Delta\eta}$.

The other line computes where a point mass can fly. Füzfa [13] modeled interstellar travel aboard the bare Kinnersley rocket, deriving trajectories, energy costs, aberration, and Doppler shifts for Proxima-bound missions, and extended the astronautics to laser-pushed sails in a companion study [14]. Those computations carry no passenger cavity, no matched shell, and none of the admissibility constraints that make the shell physical.

A systematic search — a citation scan of arXiv:2606.22531 (Semantic Scholar and INSPIRE-HEP, 2026-08-08, zero citing records), a full-text survey of v3, and an inspection of the author's public reproduction code at commit 1e9e3db [22] — found no published computation of interstellar flight profiles under the shell's admissibility constraints. The present paper performs that computation and derives its observable consequences.

This paper contributes four items. First, an admissibility-constrained maneuver catalog: a destination-free rapidity ladder, mission tables for four destinations with the crew-lifetime contour, and the seat price of carrying passengers relative to an ideal photon rocket. Second, time-optimal profiles in a three-tier constraint system, including a constraint-regime crossover at $x^* \approx 0.54$ (to our knowledge not previously quantified; Sec. 4C). Third, a signature catalog mapping every profile to observer-frame light curves, with an exact forward null during saturated acceleration, a deceleration flash governed by a general closed-form decay law, and a gravitational-wave silence discriminator. Fourth, a verification methodology (certification tests, dual-path checks, authority labels, and human decision gates over an AI workflow) whose record includes six cross-validations against the author's code and one reproducibility gap, reported factually in Sec. 3.

## 2 Framework

We use the framework of Ref. [1] as published and do not re-derive it. The exterior is the Kinnersley–Robinson–Trautman photon rocket sourced by outgoing null dust; its rest-frame exhaust pattern is the exact dipole $4\pi n^2(\vartheta) = -\dot{m} - 3ma\cos\vartheta$ (Eq. (13) of Ref. [1]), with $\vartheta$ measured from the proper-acceleration axis in the momentary rest frame. Bulk energy conditions reduce to $n^2 \geq 0$, whose forward-pole binding yields the control law $-\dot{m} \geq 3m|a|$ (Eq. (14)) and the budget $m_f/m_0 = e^{-3\int |a|\,du}$ (Eq. (15)). The static anchor satisfies the surface dominant energy condition for $x = 2m/R < 24/25$ (Eq. (25)); a regular exterior foliation caps the acceleration at the kinematic ceiling $\lambda = aR < \frac{1}{2}(1-x)$ (Eq. (40)); and finite-thickness (tangential-pressure) realizability requires the rear-pole effective compactness to satisfy $x + 2\lambda < 4/5$ (Sec. 10 of Ref. [1]).

Every number we publish inherits an authority label from the source theory, following the paper's own distinction between rigorous closed forms, numerical maps, and heuristics. Label [R] marks exact closed forms; [N] marks reliance on published numerics, used strictly inside the sampled domain; [H] marks heuristics, displayed but never used as constraints. The operative constraint system is three-tiered (Table 1): a rigorous ceiling $\min\{\frac{1}{2}(1-x), \frac{1}{2}(\frac{4}{5}-x)\}$ [R]; the effective band $\underline{g}(x)$ [N], the observer-robust frontier scan of Ref. [1] on $x \in [0.1, 0.7]$; and a floor $c_{\rm cons}(x)$, a conservative independent evaluation of the paper's rigorous lower-bound chain, carried provisionally for the reason recorded in Sec. 3. All riding profiles additionally impose the ceiling as a hard clamp.

Table 1: The three-tier constraint system on $\lambda = aR$.

| Tier | Bound on $\lambda = aR$ | Authority | Domain |
|---|---|---|---|
| Ceiling | $\min\{\frac{1}{2}(1-x),\ \frac{1}{2}(\frac{4}{5}-x)\}$ | [R] (unattainable upper limit) | $x \in (0, 4/5)$ |
| Effective | $\underline{g}(x)$, linear interpolation | [N] (no extrapolation) | $x \in [0.1, 0.7]$ |
| Floor | $c_{\text{cons}}(x)$ | [R(A3)/provisional] | $x \in (0, 4/5)$ |

# 3 Methods

## 3.1 Verification protocol

The computation is organized as a verified pipeline rather than a single code. Equations enter through a formula ledger transcribed from the source paper (v3 throughout; the v2 to v3 equation renumbering is mapped in the ledger); every implemented equation carries a certification test that reproduces a value printed in the paper before it may be used downstream. The certification suite contains eighteen gates (C1–C18), from the Tsiolkovsky budget and the surface energy-condition window through the observer-frame pattern normalization of the signature layer. Principal quantities are computed along two independent paths (closed form against independent numerics), and disagreement is a stop condition rather than a tolerance to be widened silently: the catalog carries 306 verified rows in the maneuver layer and 567 rows in the signature layer, with zero omissions.

Two path-agreement tolerances apply. Closed-form-capable rows meet $10^{-8}$, and the mission-class retarded-time maps agree to $5.0 \times 10^{-15}$. For frontier-riding profiles the two implementations of the retarded-time map agree to $1.06 \times 10^{-6}$ against a documented class tolerance of $2 \times 10^{-6}$: the stored ride grids information-limit the interpolation (kinks of the piecewise-linear $\underline{g}$ fall inside grid intervals, and the early-burn $\lambda$ gradient at $x_0 = 0.7$ is steep), so the residual reflects the consumed data, not the implementations. This class tolerance was itself ratified at a human decision gate (2026-08-09) under a project rule that tolerance changes are human-approval items.

All numerical claims in this paper, including table entries and figure captions, are machine-injected from the certified catalogs through a generated macro layer; a manifest records the source field and display transform of every injected value, and a certification test scans the manuscript for bare numeric literals and recomputes the manifest.

## 3.2 AI involvement

All computations, implementations, and manuscript drafting were performed by Claude (Fable 5, Anthropic) under a human-gated protocol: every phase deliverable was reviewed and approved at explicit human decision gates, all numerical claims are machine-injected from the certified catalogs, and tolerance changes were themselves human-gate items.

## 3.3 Cross-validation against the author's code and one gap

We pinned the author's public reproduction code [22] at commit 1e9e3db and obtained six agreements: the frontier pins $\underline{g}(x)$ match our transcription verbatim; the axial margin formula (Eq. (42) of Ref. [1]) matches token for token; the quoted worst ratio of the frozen axial threshold to the ceiling (0.9931 at $x \approx 0.631$) is reproduced independently from our own transcriptions; the stated accuracy domain of the closed-form estimate is confirmed on its stated range; the frozen-margin coefficients agree symbolically; and the code states explicitly that the v2 bifurcation scalar was removed in v3, confirming a structural retraction we had identified independently and quarantined in our harness.

Table 2: Rapidity ladder (destination-free, all [R]). The $1 - \beta$ column resolves the approach to light speed where $\beta$ saturates the displayed digits; the seat price is the initial-mass ratio relative to an ideal collimated photon rocket.

| $\eta$ | $\beta$ | $1-\beta$ | $m_f/m_0$ | radiated | seat price $e^{2\eta}$ |
|---|---|---|---|---|---|
| 0.24 | 0.2355 | $7.6 \times 10^{-1}$ | $4.9 \times 10^{-1}$ | 51.3% | $1.6 \times 10^{0}$ |
| 1 | 0.7616 | $2.4 \times 10^{-1}$ | $5.0 \times 10^{-2}$ | 95.0% | $7.4 \times 10^{0}$ |
| 3 | 0.9951 | $4.9 \times 10^{-3}$ | $1.2 \times 10^{-4}$ | 100.0% | $4.0 \times 10^{2}$ |
| 5 | 0.9999 | $9.1 \times 10^{-5}$ | $3.1 \times 10^{-7}$ | 100.0% | $2.2 \times 10^{4}$ |
| 8 | 1.0000 | $2.3 \times 10^{-7}$ | $3.8 \times 10^{-11}$ | 100.0% | $8.9 \times 10^{6}$ |
| 12 | 1.0000 | $7.6 \times 10^{-11}$ | $2.3 \times 10^{-16}$ | 100.0% | $2.6 \times 10^{10}$ |

One quantity resisted verification. The evaluated floor values of Eq. (73) could not be independently reproduced from the paper text and the public code; the three suprema they require are described procedurally in both sources but given in closed form in neither. We therefore constructed a conservative independent floor and label it provisionally. Our $c_{\text{cons}}(x)$ evaluates strictly below the printed values at all five sampled compactnesses and reproduces the printed binding-branch structure; because every step of its chain is either a printed closed form or a provable over-estimate under a documented reading, it is a true lower bound within that reading. The discrepancy is recorded as a reproducibility statement, not as an error claim, and the floor tier of every product in this paper carries the provisional label.

# 4 Results

## 4.1 The rapidity ladder and the seat price

Fuel is destination-free: $m_f/m_0 = e^{-3\Delta\eta}$ depends only on the integrated rapidity, so the first catalog table (Table 2) has no distance column. The mandatory dipole breadth of the news-silent exhaust makes the passenger shell pay $e^{3\eta}$ where an ideal collimated photon rocket pays $e^{\eta}$; the seat price, their ratio, is $e^{2\eta}$. At the worked-burn rapidity $\eta = 0.24$ the shell radiates 51.3% of its rest mass (matching the source paper's worked maneuver, which our suite reproduces as certification C4), and the seat price is a modest 1.6; at $\eta = 12$ the seat price reaches $2.6 \times 10^{10}$.

## 4.2 Mission tables and the crew-lifetime contour

Mission kinematics (hyperbolic burns, coast at fixed rapidity) convert the ladder into timetables for four destinations (Proxima Centauri, TRAPPIST-1, Sgr A*, M31; catalog distances fixed with sources in the data release). Burn durations are microseconds for kilometer-scale shells, so mission times are cruise-dominated, and the ship proper time is $\tau \approx D/\sinh\eta$. Table 3 reports the crew-lifetime contour $\tau = 50$ yr: a Proxima flyby needs only $\eta = 0.0849$ at mass ratio 0.775; Sgr A* needs $\eta = 6.947$ at $8.9 \times 10^{-10}$; an M31 flyby needs $\eta = 11.53$ at $m_f/m_0 = 9.5 \times 10^{-16}$. Arrival (deceleration into the destination system) doubles the exponent, and a round trip quadruples it: the M31 arrival costs $9.1 \times 10^{-31}$, which prices the mission rather than forbidding it, since the budget is positive-energy throughout. For round trips the lifetime condition applies to the total out-and-back proper time, so the round-trip column of Table 3 is evaluated at its own cruise rapidity $\eta_{\text{RT}} = \operatorname{arcsinh}(2D/\tau)$ (for M31, $\eta_{\text{RT}} = 12.22$), and its fuel column uses $\Delta\eta_{\text{tot}} = 4\eta_{\text{RT}}$. Figure 1 shows the contour across the ladder.

Table 3: Crew-lifetime contour ($\tau = 50$ yr, ship proper time; burns included exactly, cruise-dominated). The flyby and arrival columns share $\eta_{50} = \text{arcsinh}(D/\tau)$; the round-trip column is evaluated at its own $\eta_{\text{RT}} = \text{arcsinh}(2D/\tau)$, the rapidity at which the total out-and-back proper time meets the contour. Fuel columns are $m_f/m_0 = e^{-3\Delta\eta_{\text{tot}}}$ with $\Delta\eta_{\text{tot}} = \eta_{50}, 2\eta_{50}, 4\eta_{\text{RT}}$.

| Destination | $D$ [ly] | $\eta_{50}$ | $\eta_{\text{RT}}$ | $m_f/m_0$ (flyby) | (arrive) | (round trip) |
|---|---|---|---|---|---|---|
| Proxima Centauri | 4.25 | 0.085 | 0.169 | 7.8e-01 | 6.0e-01 | 1.3e-01 |
| TRAPPIST-1 | 40.5 | 0.740 | 1.260 | 1.1e-01 | 1.2e-02 | 2.7e-07 |
| Sgr A* | $2.6 \times 10^4$ | 6.947 | 7.640 | 8.9e-10 | 7.9e-19 | 1.5e-40 |
| M31 (Andromeda) | $2.54 \times 10^6$ | 11.529 | 12.222 | 9.5e-16 | 9.1e-31 | 2.0e-64 |

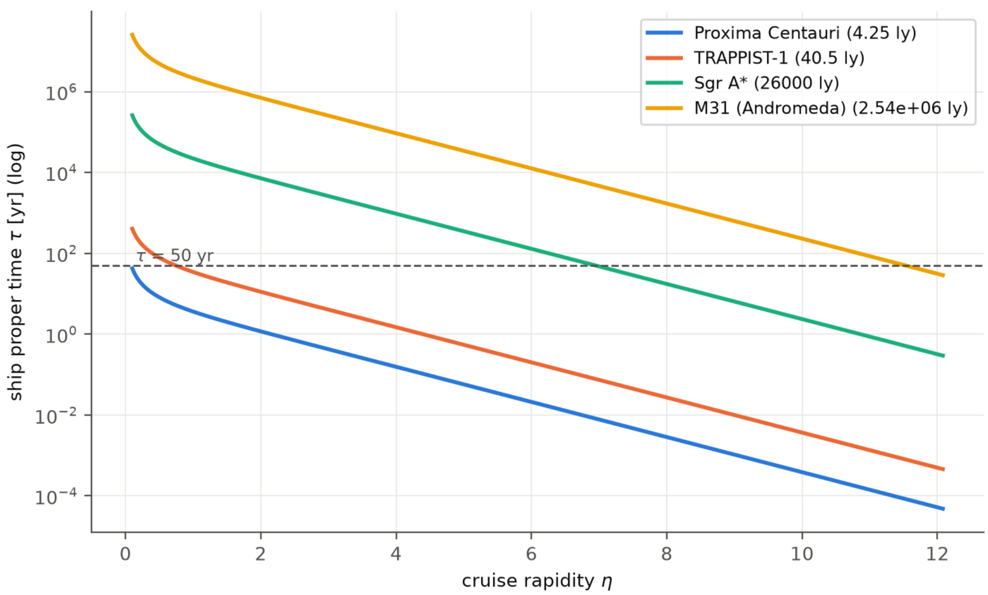


Figure 1: Ship proper time versus cruise rapidity for arrival missions to the four catalog destinations, with the 50-year contour marked. Crossings occur at the $\eta_{50}$ values of Table 3.

### 4.3 Three-tier time-optimal profiles and the constraint-regime crossover

The time-optimal maneuver at fixed $\Delta\eta$ saturates the frontier almost everywhere (Theorem 8 of Ref. [1]), so the minimum time is the quadrature $T = R\int_0^{\Delta\eta} d\eta'/\lambda_{\text{tier}}(x_0 e^{-3\eta'})$ along the moving wall: the shell radiates, $x$ falls, and the wall widens during the burn. We evaluate this riding time in all three tiers (Fig. 3, Table 4). The effective [N] band leaves its sampled domain at $\eta_{\text{fb}} = \ln(x_0/0.1)/3 = 0.366$ (for $x_0 = 0.3$) and falls back to the rigorous ceiling, with the fallback point recorded in every profile; for $\Delta\eta \gtrsim 1$ the effective and ceiling times converge, so the bracket tightens automatically. The floor tier is slower by orders of magnitude and grows as $e^{3\Delta\eta}$; this is the conservatism of a proof, not a property of nature, and in SI units even the floor-tier burn at $\Delta\eta = 0.24$ lasts milliseconds for $R = 1$ km.

The three tiers also locate a structural fact about the constraint stack (Fig. 2). The thin-shell frontier scan $\underline{g}(x)$ [N] crosses the thick-wall ceiling $(4/5 - x)/2$ [R] at $x^* \approx 0.54$ ($\lambda^* \approx 0.13$, under the linear interpolation of the published pins), a crossover which has not, to our knowledge, been quantified before (search scope in Sec. 1): below $x^*$ the dominant-energy frontier binds, above it

Table 4: Frontier-riding minimum times at $x_0 = 0.3$ (units of $R$; multiply by $R/c$ for SI). The ceiling column is an unattainable lower bound (the ceiling is an open condition); the fallback column marks where the [N] band exits its sampled domain.

| $\Delta\eta$ | $T_{\text{floor}}/R$ | $T_{\text{eff}}/R$ | $T_{\text{ceil}}/R$ | $\eta_{\text{fb}}$ |
|---:|---:|---:|---:|---:|
| 0.1 | 1e+03 | 0.515 | 0.37 | – |
| 0.24 | 2.58e+03 | 1.22 | 0.824 | – |
| 0.5 | 6.88e+03 | 2.25 | 1.57 | 0.366 |
| 1 | 3.01e+04 | 3.56 | 2.88 | 0.366 |
| 2 | 5.65e+05 | 6.08 | 5.39 | 0.366 |
| 3 | 1.12e+07 | 8.58 | 7.89 | 0.366 |
| 5 | 4.53e+09 | 13.6 | 12.9 | 0.366 |
| 8 | 3.67e+13 | 21.1 | 20.4 | 0.366 |
| 12 | 5.98e+18 | 31.1 | 30.4 | 0.366 |

Table 5: SI conversions at $x_0 = 0.3$ (masses scale as $R$; the peak luminosity $\frac{3}{2}x\lambda\,(c^5/G)$ is $R$-independent).

| $R$ [m] | $m$ [kg] | $m$ [$M_\odot$] | $L_{\text{peak}}$ [W] | [$L_\odot$] |
|---:|---:|---:|---:|---:|
| 100 | 2.02e+28 | 0.0102 | 3.10e+51 | 8.10e+24 |
| 1000 | 2.02e+29 | 0.102 | 3.10e+51 | 8.10e+24 |
| 10000 | 2.02e+30 | 1.02 | 3.10e+51 | 8.10e+24 |

thick-wall realizability binds, and thin-shell catalog values beyond $x^*$ are reachable only in the distributional idealization. The physical content is consistent with the source paper's own nesting of admissibility windows; the crossover simply makes the switch of operative regime quantitative.

### 4.4 The SI layer

The dimensionless catalog converts to SI through a single certified module. At $x_0 = 0.3$ and $R = 1$ km the shell mass is $2.02 \times 10^{29}$ kg $= 0.102\,M_\odot$; the saturated riding luminosity is $L = \frac{3}{2}x\lambda\,(c^5/G) = 3.1 \times 10^{51}$ W $= 8.1 \times 10^{24}\,L_\odot$, independent of $R$; and one unit of rapidity takes $17.6\,\mu$s of shell time (Table 5). A planet-to-star-mass object burns brighter than any supernova for microseconds at a time. The luminosity is the price the conservation law charges for steering, paid as positive radiation.

### 4.5 Observational signatures: the forward null and the deceleration flash

Every maneuver profile maps to observer-frame light curves through the exact rest-frame pattern and special-relativistic aberration, Doppler boost, and retarded-time compression, reusing the mechanics of Füzfa [13]. Three facts organize the catalog (Figs. 4–6).

First, a shell accelerating at the saturated control law is exactly dark at its forward pole: $4\pi n^2 = 3ma(1 - \cos\vartheta)$ vanishes at $\vartheta = 0$, aberration fixes the poles, and our certification verifies the null in exact rational arithmetic. An approaching shell is invisible head-on for the entire boost phase.

Second, deceleration reverses the lobe: the destination observer receives the exhaust face-on, boosted by $\delta^4 = e^{4\eta}$ and compressed in arrival time by $dt_{\text{obs}}/du = e^{-\eta}$. Along a saturated deceleration leg the received flux follows the closed-form decay law $F \propto e^{7\eta - 3\Delta\eta_{\text{tot}}}$, obtained by

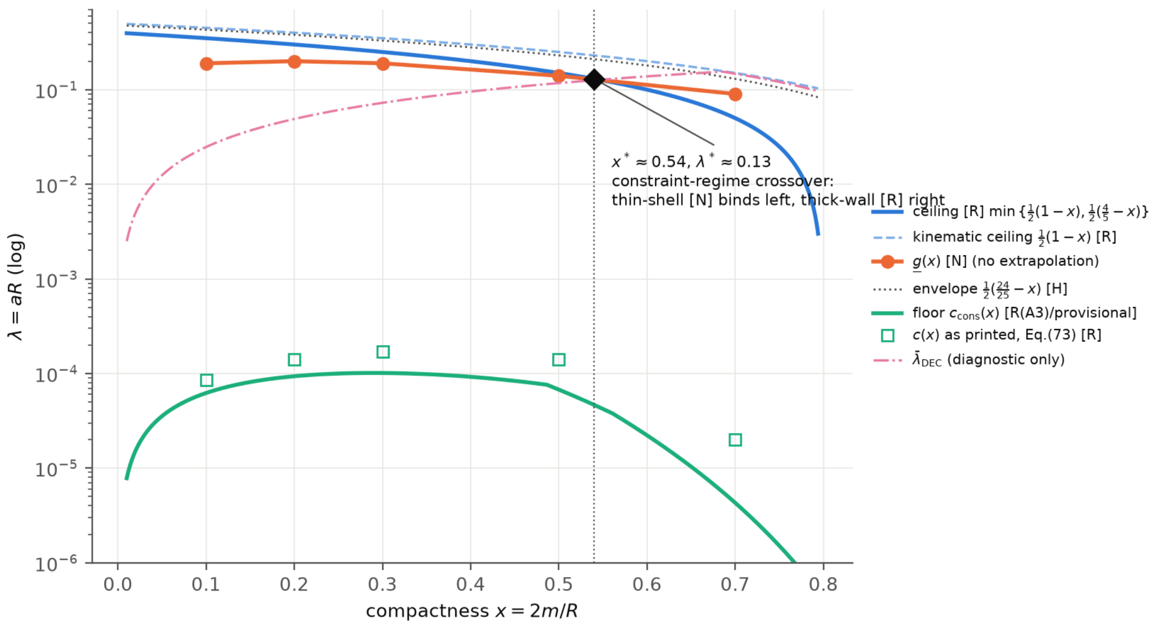


Figure 2: The three-tier constraint system on $\lambda = aR$, with the printed floor values of Eq. (73) (open squares), our conservative floor (solid), the effective band (circles), the ceilings, the heuristic envelope [H], and the frozen-shape diagnostic (dash-dotted, never used as a constraint). The diamond marks the constraint-regime crossover $x^* \approx 0.54$.

combining the boost with the Tsiolkovsky mass history; for the M31 arrival ($\Delta\eta_{\mathrm{tot}} = 24$) this is $e^{7\eta-72}$, the peak $FD^2$ is $1.48\times10^4$ in geometric units (the catalog value agrees with the closed form to better than $10^{-12}$), and the decay scale is $e^{-\eta}/(7\lambda) \approx 5 \times 10^{-6}\,R$, about 15 ps for $R = 1$ km. Low-rapidity arrivals stretch to microseconds. The arrival-type mission is therefore a matched pair: a departure-side burst at $t \approx 0$ and a destination-side flash one light-crossing of the route later.

Third, the energy closure that stitches the two layers is exact: the integrated light curve of every profile equals $m_0 - m_f$ to $10^{-10}$ (certification C15), so the timetable and the light curve are two readings of the same integral.

### 4.6 The gravitational-wave silence discriminator

The minimum-radiation maneuver is the Damour dipole, and on that maneuver the exterior radiates no gravitational waves: the Bondi news vanishes identically because the spin-raising operator annihilates the $\ell \leq 1$ emission, a property established for photon rockets by Damour [16] and holding exactly for the news-silent class of Ref. [1]. Departures from the pure dipole radiate: the news energy is a positive quadratic form in the $\ell \geq 2$ leakage amplitudes, so gravitational-wave output grows quadratically while the fuel saving from sharper collimation grows only linearly. The model therefore makes a falsifiable joint prediction, a photometric transient with no gravitational-wave counterpart at the same position and time, and the same pair of channels acts as a purity meter for the maneuver itself. The contrast with warp-drive collapse is sharp: containment failure of an Alcubierre-class bubble emits a gravitational-wave burst as its primary signal [20], the opposite corner of the discriminant plane. Proposals to search for emissions from positive-energy warp bubbles [21] can draw the required waveforms and angular patterns directly from the present

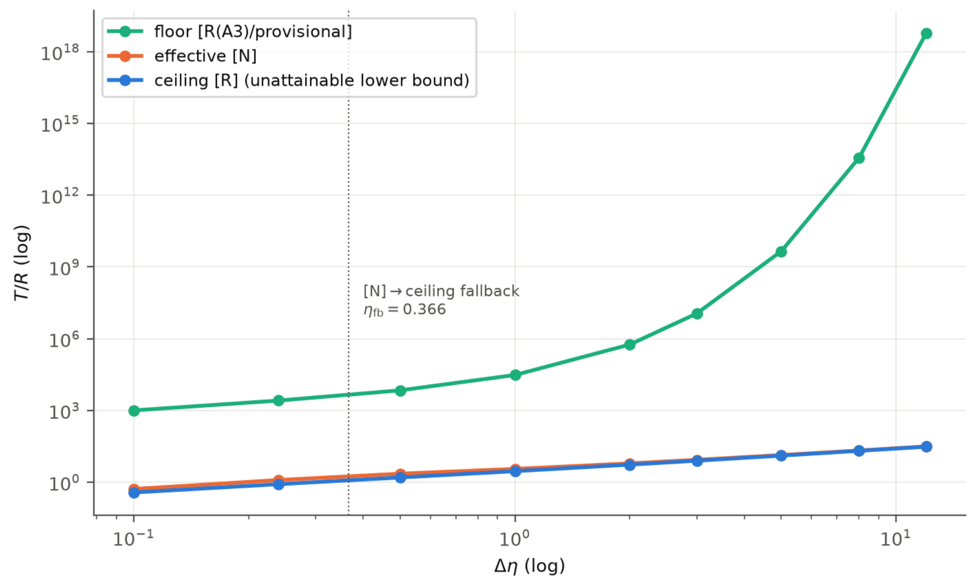


Figure 3: Minimum riding time versus $\Delta\eta$ at $x_0 = 0.3$ in the three tiers. The effective and ceiling curves converge beyond the fallback point; the floor grows as $e^{3\Delta\eta}$.

catalog.

# 5 Discussion

## 5.1 Limitations

The catalog's validity boundaries are carried by its labels. The accelerated thick-wall energy-condition theorem is open in the source theory (its author states it as future work), so thick-wall realizability is imposed as the static window on the rear-pole effective compactness. The effective band $\underline{g}(x)$ is defined only on $x \in [0.1, 0.7]$ and is never extrapolated; riding profiles fall back to the rigorous ceiling outside it, with fallback points recorded. Signatures are bolometric with parameterized scenarios only (a blackbody reading of the $R = 1$ km shell would peak in hard gamma rays, but that is a labeled assumption, not a claim about the exhaust microphysics). The floor tier remains provisional pending the reconciliation described in Sec. 3. Interpretive assumptions introduced by this work (the dimensionless closure of the worked burn, the mission burn convention, the floor-chain reading, and the thin-shell clamp rationale) are enumerated in the assumptions ledger of the data release.

## 5.2 Position in the time–luminosity plane

The predicted transients occupy a corner of the time–luminosity plane that known astrophysical classes do not: microsecond-to-picosecond durations at equivalent isotropic luminosities far above supernova peaks, in single events or matched pairs, with an exact forward null before arrival and gravitational silence throughout. We make no statement about detectability with any instrument; the physical geometry (beamed within $\sim e^{-\eta}$, non-repeating, paired across a light-crossing of the route) suffices to distinguish the class from repeating or gravitationally loud phenomena at the level of scaling laws.

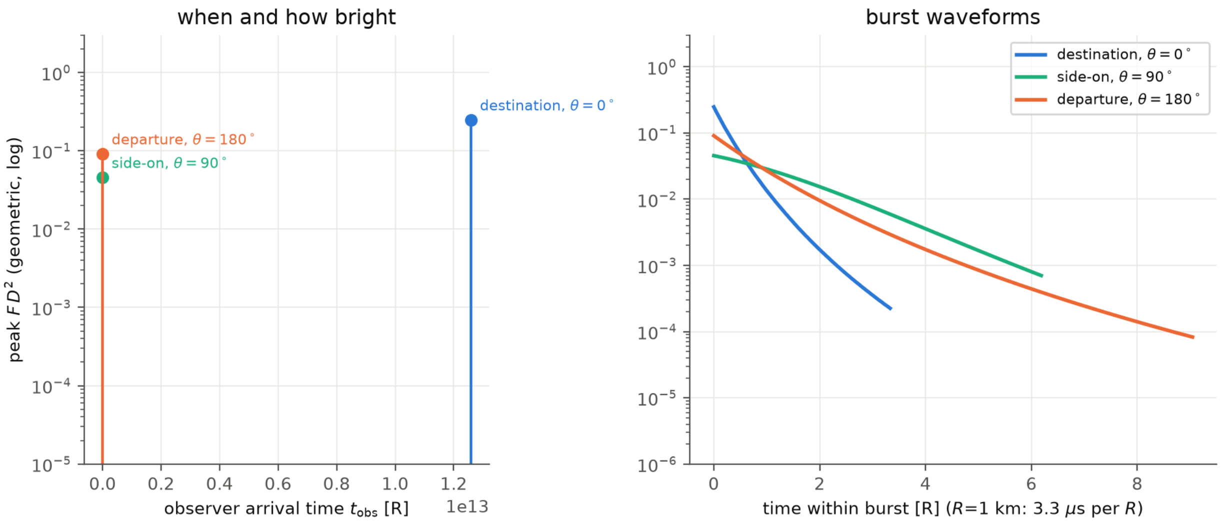


Figure 4: Arrival-mission fingerprint (Proxima, $\eta = 1$). Left: arrival schedule of the transients seen by the three canonical observers. Right: burst waveforms; the destination flash is the brightest and shortest.

### 5.3 Reproducibility of the floor values

We restate the one open verification item as an invitation. The printed floor evaluations of Eq. (73) require three suprema whose closed forms appear in neither the paper nor the public code at the pinned commit; our conservative reconstruction bounds them from above under a documented reading and lands strictly below the printed values. Publication of the computation behind Eq. (73) would close the gap and likely sharpen the floor tier of this catalog.

## 6 Data availability

The full pipeline (certified library, certification suite C1–C19, catalogs, signature files, figure and table generators, the numbers manifest, and the assumptions ledger) is released at [GitHub URL, Zenodo DOI to be minted at release], pinned at [commit hash injected at release] of the project repository. The verification suite runs in under one minute and reproduces every number in this paper.

## Acknowledgments

This work uses warpax [23] and the world_tube reproduction code [22], both MIT-licensed, by An T. Le. All computations, implementations, and drafting were performed by Claude (Fable 5, Anthropic) under the human-gated protocol described in Sec. 3; the human author directed the project, arbitrated all decision gates, and bears responsibility for the content.

## References

[1] A. T. Le, "Steering a warp drive without exotic matter," arXiv:2606.22531 (2026).

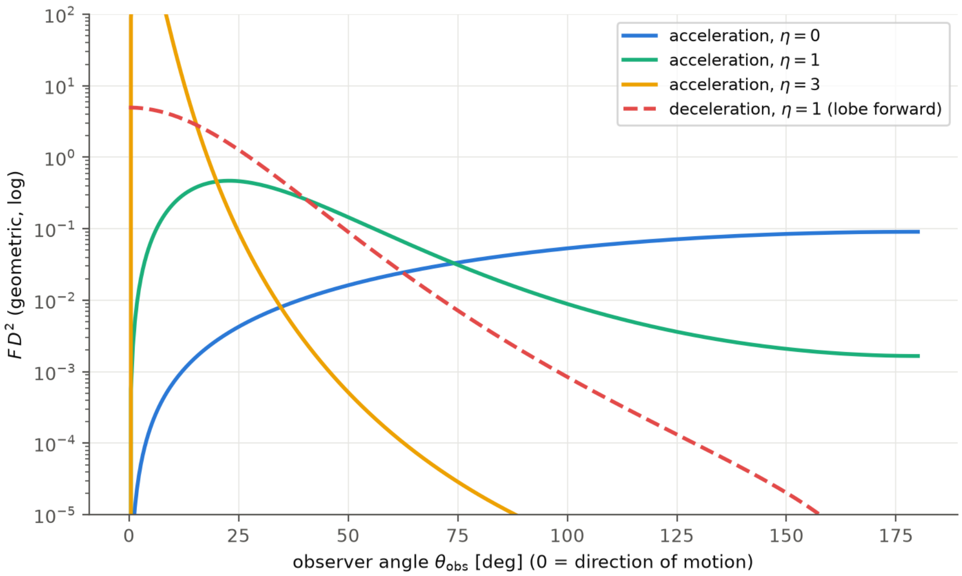


Figure 5: Observer-frame angular pattern of a saturated burn at several rapidities, and the reversed lobe of a deceleration burn. The forward null of the acceleration pattern is exact.

[2] A. T. Le, "On the boundary cost of source-consistent warp shells," arXiv:2605.25417 (2026).

[3] A. T. Le, "Observer-robust energy condition verification for warp drive spacetimes," arXiv:2602.18023 (2026).

[4] M. Alcubierre, Class. Quantum Grav. **11**, L73 (1994).

[5] J. Santiago, S. Schuster, and M. Visser, Phys. Rev. D **105**, 064038 (2022).

[6] M. J. Pfenning and L. H. Ford, Class. Quantum Grav. **14**, 1743 (1997).

[7] F. S. N. Lobo and M. Visser, Class. Quantum Grav. **21**, 5871 (2004).

[8] E. W. Lentz, Class. Quantum Grav. **38**, 075015 (2021).

[9] B. Celmaster and S. Rubin, arXiv:2511.18251 (2025).

[10] A. Bobrick and G. Martire, Class. Quantum Grav. **38**, 105009 (2021).

[11] J. Fuchs, C. Helmerich, A. Bobrick, L. Sellers, B. Melcher, and G. Martire, Class. Quantum Grav. **41**, 095013 (2024).

[12] C. Helmerich, J. Fuchs, A. Bobrick, L. Sellers, B. Melcher, and G. Martire, Class. Quantum Grav. **41**, 095009 (2024).

[13] A. Füzfa, Phys. Rev. D **99**, 104081 (2019).

[14] A. Füzfa, W. Dhelonga-Biarufu, and O. Welcomme, Phys. Rev. Research **2**, 043186 (2020).

[15] W. Kinnersley, Phys. Rev. **186**, 1335 (1969).

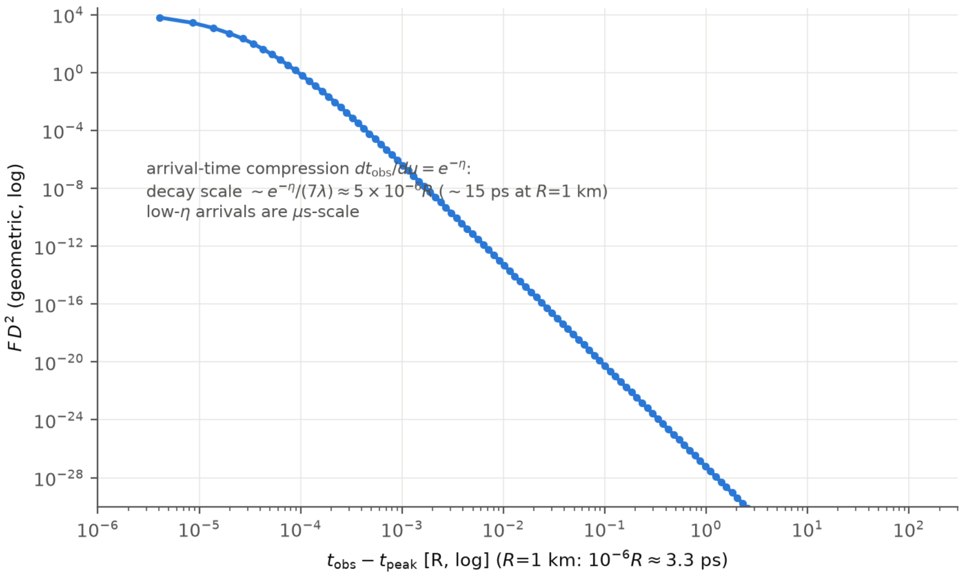


Figure 6: The M31 deceleration flash seen from the destination (logarithmic time from peak). Arrival-time compression turns a $63R$-long burn into a picosecond-scale transient.


[16] T. Damour, Class. Quantum Grav. **12**, 725 (1995), arXiv:gr-qc/9412063.

[17] W. B. Bonnor, Class. Quantum Grav. **11**, 2007 (1994).

[18] U. von der Gönna and D. Kramer, Class. Quantum Grav. **15**, 215 (1998).

[19] W. Israel, Nuovo Cimento B **44**, 1 (1966).

[20] K. Clough, T. Dietrich, and S. Khan, Open J. Astrophys. **7** (2024), arXiv:2406.02466.

[21] E. W. Lentz and R. C. Felton, arXiv:2405.19381 (2024).

[22] A. T. Le, world_tube reproduction code, https://github.com/anindex/world_tube (MIT license).

[23] A. T. Le, warpax v1.3.0, https://github.com/anindex/warpax (MIT license).